DEAKIN UNIVERSITY | DEAKIN APPLIED ARTIFICIAL INTELLIGENCE INITIATIVE

Extraction of Chemical Synthesis Pathways using LLMs

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Background and problem statement

The synthesis of pharmaceutical compounds is a cornerstone of drug discovery and chemical innovation. Over decades, academic researchers have published millions of papers describing synthetic routes, reaction conditions, yields, catalysts, solvents, and intermediate structures. This information is predominantly embedded in unstructured natural language, figures, reaction schemes, and supplementary materials scattered across journals and publishers. While this collective body of knowledge is extraordinarily rich, it remains difficult to systematically access, compare, and reuse at scale.

Traditionally, the extraction of drug synthesis pathways from the literature has relied on manual curation by expert chemists or on narrowly scoped rule-based text-mining systems. Recent advances in large language models (LLMs) offer a promising alternative. Despite their promise, LLM-based approaches to extracting drug synthesis pathways face several unresolved challenges. Academic chemistry papers rarely present synthesis routes in a single, explicit, step-by-step format. Instead, pathways are distributed across sections of text, tables, reaction schemes, footnotes, and supplementary information. Critical details are often implicit and require domain reasoning to reconstruct. Furthermore, chemical entities may be referenced inconsistently, using trivial names, abbreviations, or structural descriptions that vary across papers.

The core problem addressed by this internship project is therefore: how to reliably extract complete, structured and chemically meaningful drug synthesis pathways from unstructured academic papers using large language models. This includes identifying individual reaction steps, mapping reactants to products, capturing reagents and conditions, resolving references across the document, and assembling these elements into coherent multi-step synthesis graphs.

Aim

To curate consistent, structured drug synthesis pathway data from academic chemistry papers using LLMs and to build a reliable database that enables systematic analysis and reuse of published synthetic knowledge.

Objectives

- To design and apply LLM-based extraction pipelines that identify and standardise drug synthesis steps, reagents, conditions, and intermediates from unstructured academic papers.
- To construct and validate a structured database of curated synthesis pathways, ensuring consistency, traceability to source literature, and chemical plausibility.

Expected outcomes

- A validated, searchable database of consistently curated drug synthesis pathways extracted from academic literature, demonstrating the feasibility of using LLMs for scalable and chemically meaningful knowledge extraction.

How will this opportunity benefit you?

- Basic understanding of LLMs, experience in a pharmaceutical science.

Other details

- Located on-campus at Burwood
- 10 weeks in duration (Unpaid)
- Undertaken as 2 days per week for a total of ~160 hours
- To be completed within the 12-week trimester period